

Analysis First-Year Exam, August 2018, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let the sequence $(s_n)_{n=1}^\infty$ in $[0, 1]$ be uniformly distributed in the sense that if $0 \leq a \leq b \leq 1$ then

$$\lim_{n \rightarrow \infty} \frac{\#\{k \leq n : s_k \in [a, b]\}}{n} = b - a.$$

Let $f : [0, 1] \rightarrow \mathbb{R}$ and prove that

$$\lim_{n \rightarrow \infty} \frac{f(s_1) + \cdots + f(s_n)}{n} = \int_0^1 f(t) dt$$

in each of the following cases:

- (i) f is a step function (a finite linear combination of indicators of intervals);
 - (ii) f is Riemann-integrable.
2. Fix $a \in \mathbb{R}$ and for each integer $n > 0$ write $f_n(t) = n^a t(1 - t^2)^n$ whenever $0 \leq t \leq 1$.
- (i) Show that $(f_n)_{n=1}^\infty$ converges pointwise on $[0, 1]$; say to f .
 - (ii) For which values of a does $(f_n)_{n=1}^\infty$ converge uniformly on $[0, 1]$? Justify.
 - (iii) For which values of a is it true that $\int_0^1 f_n \rightarrow \int_0^1 f$? Justify.
3. Let $f : [1, \infty) \rightarrow \mathbb{R}$ be continuous and satisfy $\lim_{t \rightarrow \infty} f(t) = A \in \mathbb{R}$. Prove that there is a sequence $(p_n)_{n=0}^\infty$ of polynomials such that $p_n(1/t)$ converges to $f(t)$ uniformly for $t \geq 1$.
4. Let $(f_n)_{n=1}^\infty$ be a sequence of real-valued measurable functions. Prove that each of the following sets is measurable:
- (i) $B = \{\omega : (f_n(\omega))_{n=1}^\infty \text{ has no biggest term}\}$;
 - (ii) $C = \{\omega : \cos(f_n(\omega)) > 0 \text{ for each } n > 0\}$;
 - (iii) $D = \{\omega : (f_n(\omega))_{n=1}^\infty \text{ does not converge to a rational number}\}$.
5. State the Monotone Convergence Theorem. Use it to prove the following: let $(f_n)_{n=1}^\infty$ be a sequence of non-negative integrable functions with pointwise limit f ; if $\int f_n d\mu \leq M < \infty$ for each n then f is integrable and $\int f d\mu \leq M$.